

DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis

Birgit Pfitzmann
IBM Research
Rueschlikon, Switzerland
bpf@zurich.ibm.com

Christoph Auer
IBM Research
Rueschlikon, Switzerland
cau@zurich.ibm.com

Michele Dolfi
IBM Research
Rueschlikon, Switzerland
dol@zurich.ibm.com

Ahmed S. Nassar
IBM Research
Rueschlikon, Switzerland
ahn@zurich.ibm.com

Peter Staar
IBM Research
Rueschlikon, Switzerland
taa@zurich.ibm.com

ABSTRACT

Accurate document layout analysis is a key requirement for high-quality PDF document conversion. With the recent availability of public, large ground-truth datasets such as PubLayNet and DocBank, deep-learning models have proven to be very effective at layout detection and segmentation. While these datasets are of adequate size to train such models, they severely lack in layout variability since they are sourced from scientific article repositories such as PubMed and arXiv only. Consequently, the accuracy of the layout segmentation drops significantly when these models are applied on more challenging and diverse layouts. In this paper, we present *DocLayNet*, a new, publicly available, document-layout annotation dataset in COCO format. It contains 80863 manually annotated pages from diverse data sources to represent a wide variability in layouts. For each PDF page, the layout annotations provide labelled bounding-boxes with a choice of 11 distinct classes. *DocLayNet* also provides a subset of double- and triple-annotated pages to determine the inter-annotator agreement. In multiple experiments, we provide baseline accuracy scores (in mAP) for a set of popular object detection models. We also demonstrate that these models fall approximately 10% behind the inter-annotator agreement. Furthermore, we provide evidence that *DocLayNet* is of sufficient size. Lastly, we compare models trained on PubLayNet, DocBank and *DocLayNet*, showing that layout predictions of the *DocLayNet*-trained models are more robust and thus the preferred choice for general-purpose document-layout analysis.

CCS CONCEPTS

• **Information systems** → **Document structure**; • **Applied computing** → **Document analysis**; • **Computing methodologies** → **Machine learning**; **Computer vision**; **Object detection**;

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

KDD '22, August 14–18, 2022, Washington, DC, USA

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9385-0/22/08.

<https://doi.org/10.1145/3534678.3539043>

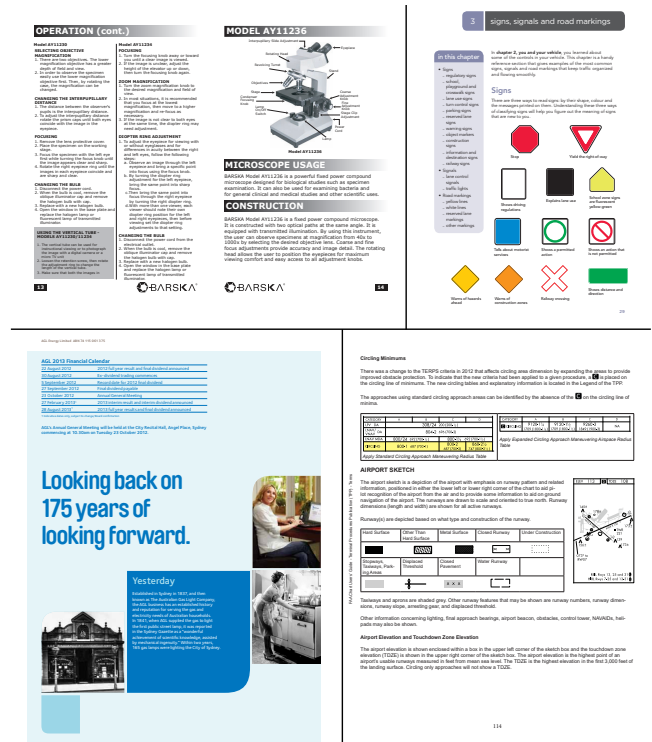


Figure 1: Four examples of complex page layouts across different document categories

KEYWORDS

PDF document conversion, layout segmentation, object-detection, data set, Machine Learning

ACM Reference Format:

Birgit Pfitzmann, Christoph Auer, Michele Dolfi, Ahmed S. Nassar, and Peter Staar. 2022. DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22)*, August 14–18, 2022, Washington, DC, USA. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3534678.3539043>

Table 1: DocLayNet dataset overview. Along with the frequency of each class label, we present the relative occurrence (as % of row “Total”) in the train, test and validation sets. The inter-annotator agreement is computed as the mAP@0.5-0.95 metric between pairwise annotations from the triple-annotated pages, from which we obtain accuracy ranges.

class label	Count	% of Total			triple inter-annotator mAP @ 0.5-0.95 (%)						
		Train	Test	Val	All	Fin	Man	Sci	Law	Pat	Ten
Caption	22524	2.04	1.77	2.32	84-89	40-61	86-92	94-99	95-99	69-78	n/a
Footnote	6318	0.60	0.31	0.58	83-91	n/a	100	62-88	85-94	n/a	82-97
Formula	25027	2.25	1.90	2.96	83-85	n/a	n/a	84-87	86-96	n/a	n/a
List-item	185660	17.19	13.34	15.82	87-88	74-83	90-92	97-97	81-85	75-88	93-95
Page-footer	70878	6.51	5.58	6.00	93-94	88-90	95-96	100	92-97	100	96-98
Page-header	58022	5.10	6.70	5.06	85-89	66-76	90-94	98-100	91-92	97-99	81-86
Picture	45976	4.21	2.78	5.31	69-71	56-59	82-86	69-82	80-95	66-71	59-76
Section-header	142884	12.60	15.77	12.85	83-84	76-81	90-92	94-95	87-94	69-73	78-86
Table	34733	3.20	2.27	3.60	77-81	75-80	83-86	98-99	58-80	79-84	70-85
Text	510377	45.82	49.28	45.00	84-86	81-86	88-93	89-93	87-92	71-79	87-95
Title	5071	0.47	0.30	0.50	60-72	24-63	50-63	94-100	82-96	68-79	24-56
Total	1107470	941123	99816	66531	82-83	71-74	79-81	89-94	86-91	71-76	68-85

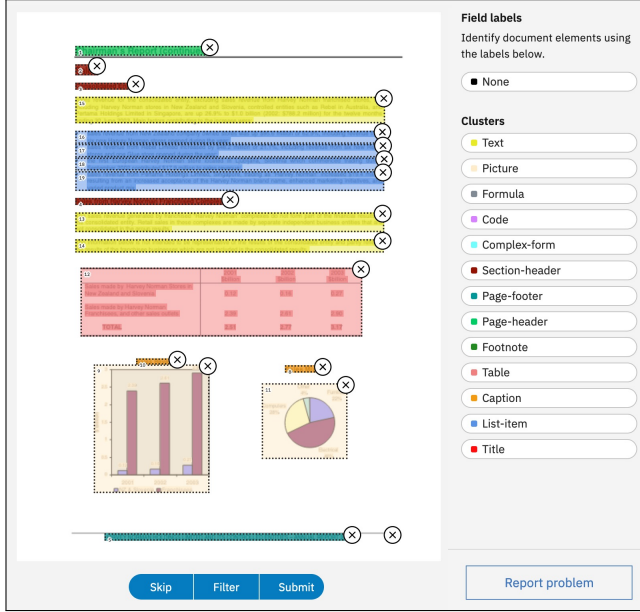


Figure 3: Corpus Conversion Service annotation user interface. The PDF page is shown in the background, with overlaid text-cells (in darker shades). The annotation boxes can be drawn by dragging a rectangle over each segment with the respective label from the palette on the right.

we distributed the annotation workload and performed continuous quality controls. Phase one and two required a small team of experts only. For phases three and four, a group of 40 dedicated annotators were assembled and supervised.

Phase 1: Data selection and preparation. Our inclusion criteria for documents were described in Section 3. A large effort went into ensuring that all documents are free to use. The data sources

include publication repositories such as arXiv³, government offices, company websites as well as data directory services for financial reports and patents. Scanned documents were excluded wherever possible because they can be rotated or skewed. This would not allow us to perform annotation with rectangular bounding-boxes and therefore complicate the annotation process.

Preparation work included uploading and parsing the sourced PDF documents in the Corpus Conversion Service (CCS) [22], a cloud-native platform which provides a visual annotation interface and allows for dataset inspection and analysis. The annotation interface of CCS is shown in Figure 3. The desired balance of pages between the different document categories was achieved by selective subsampling of pages with certain desired properties. For example, we made sure to include the title page of each document and bias the remaining page selection to those with figures or tables. The latter was achieved by leveraging pre-trained object detection models from PubLayNet, which helped us estimate how many figures and tables a given page contains.

Phase 2: Label selection and guideline. We reviewed the collected documents and identified the most common structural features they exhibit. This was achieved by identifying recurrent layout elements and lead us to the definition of 11 distinct class labels. These 11 class labels are *Caption*, *Footnote*, *Formula*, *List-item*, *Page-footer*, *Page-header*, *Picture*, *Section-header*, *Table*, *Text*, and *Title*. Critical factors that were considered for the choice of these class labels were (1) the overall occurrence of the label, (2) the specificity of the label, (3) recognisability on a single page (i.e. no need for context from previous or next page) and (4) overall coverage of the page. Specificity ensures that the choice of label is not ambiguous, while coverage ensures that all meaningful items on a page can be annotated. We refrained from class labels that are very specific to a document category, such as *Abstract* in the *Scientific Articles* category. We also avoided class labels that are tightly linked to the semantics of the text. Labels such as *Author* and *Affiliation*, as seen in DocBank, are often only distinguishable by discriminating on

³<https://arxiv.org/>